



# Pestilential impacts of battery industry discharged metal waste on human health

Sahil Randhawa, Lalita Chopra \*

University Institute of Sciences (UIS), Department of Chemistry, Chandigarh University, Gharuan, Mohali, Punjab 140413, India

## ARTICLE INFO

### Article history:

Available online 21 September 2021

### Keywords:

Life-cycle impact assessment (LCIA)

Lithium batteries

Toxicity

Hazardous

Li-ion batteries

## ABSTRACT

Lithium-ion and lithium-polymer batteries have become dominant in consumer electronic products because of their energy density and product longevity. Li dry-cell can give significantly to sustainable waste as well as potential pollutants that are harmful to human health due to their small size, high rates of removal from buy item here whatever we are previously owned, as well as a, require of the same rigid plan at inherent discarding. To assess hazard assessment, source reduction capability, and dangerous potential about lithium battery worn inside the digital telephone, we used standardized leaching monitoring, Lifecycle Impact Assessment (LCIA), and Hazardous Waste Classification. Diffuse Leon batteries are listed as dangerous by US federal regulations due to their main contents. However, due to high levels of cobalt, copper, and nickel, all lithium batteries tested in California are rated as hazardous. Leached amounts of chromium, lead, and thallium in some of Leon's batteries violated California regulations. However, due to high amounts of cobalt, copper, and nickel, all lithium batteries tested in California are rated as hazardous. Leached amounts of chromium, lead, and thallium in some of Leon's batteries violated California regulations materials.

Copyright © 2022 Elsevier Ltd. All rights reserved.

Selection and peer-review under responsibility of the scientific committee of the International Conference on Smart and Sustainable Developments in Materials, Manufacturing and Energy Engineering

## 1. Introduction

Heavy metals are distinct metals and metal developments that can have a dangerous effect on human health. Lithium, Chromium, Lead, Cobalt, Copper, Nickel are six prevailing heavy metals. These are all naturally occurring substances which are often present in the environment at low levels. In larger amounts, they can be dangerous. Ingestion or breathing is the most corporate way that humans are showing to these metals. In work at or living nearby an industrial site that expenditure these metals and their mixtures, as well as living proximate a site where these metals have been improperly expected of increases one's risk of acquaintance. Upper costs can be completed by maintenance lifestyles. [Fig. 1.](#)

**Abbreviations:** WET, Waste Extraction Test; MRCVS, Midrange Calibration Verification Standard; STLC, Soluble Threshold Limit Concentration; CB, Calibration Blank; LCS, Laboratory Control Standard illustrate.

\* Corresponding author.

E-mail address: [lalita.chemistry@cumail.in](mailto:lalita.chemistry@cumail.in) (L. Chopra).

### 1.1. Heavy metals

#### • Lithium:

Lithium batteries are also used in hybrid vehicles, military applications, and aerospace applications due to these advantages. As a result, the global lithium battery demand is expected to increase in 2014[1]. Electrically lithium batteries in movable electronic devices waste are the best rising division in 2009 by Ogunseitan O et al. [2,3]. We concentrated our research on electrically lithium batteries that use Li compound at the same time as electrode matter rather than the short-lived button-type battery with the purpose of use tinny. The improved efficient composure about cell phones has increased these markets in favor of little and elevated energy capacity lithium dry cell. As a result, everybody classified smart phone dry-cell the same as a take apart category in its search. Lithium batteries are likely to contain toxic materials in 2008 by Thomas HR et al. [4]. Although there is no consensus about what happens to damaged lithium batteries in international e-waste exposure to the synthetic substance is characteristically controlled for the duration of Li dry-cell manufacturing by realistic strength and protection law, and prospective combustion hazard

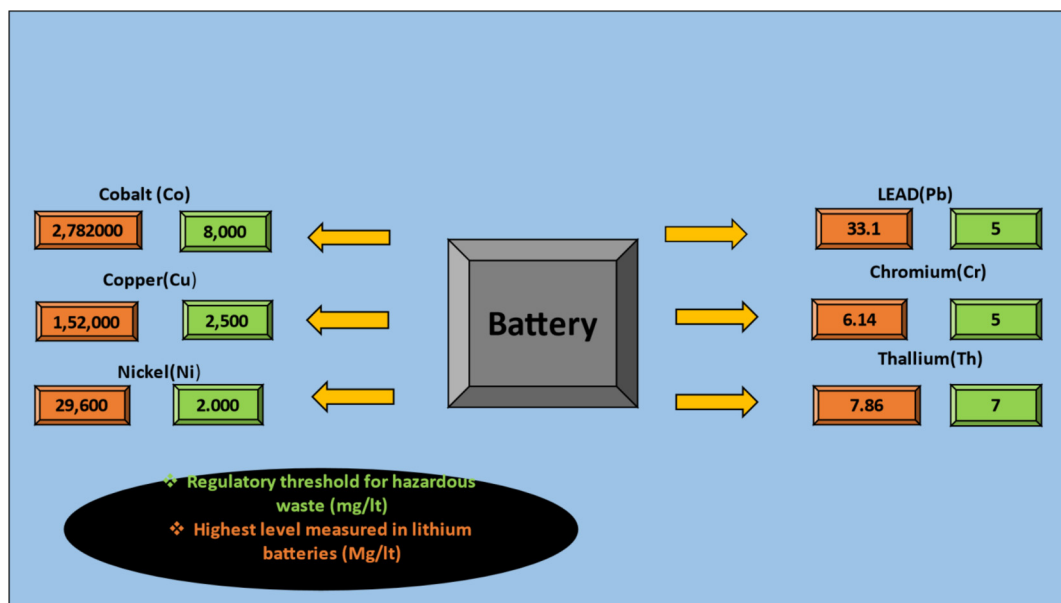


Fig. 1. Battery.

related among its transport be governed by their United States system central set of laws in 2005 by Schoenung J et al. [5,6].

According to previous reports, conventionally decommissioned mobile phones are classified as dangerous waste in central rule in 2007 by Lincoln J et al. [7,8]. Those studies, on the other hand, avoided using batteries, relied on lithium equipment, or relied on reserve model data instead of analytically defining synthetic material combinations in alternative natural discarding in 2011 by Majeau-Bettez G et al. [9-13]. As a result, the study's goals are as follows:

- 1) Using the Toxicity Characteristics leaching method, assess the metal substance of redundant electrical lithium dry-cell to might be disposed of as dangerous wastage under US central set of laws (TCLP) in 2020 by Vann k et al. [14,18,21,22] and below California state arrangement, below the Total Threshold Limit Concentrations (TCLP) [15] as well as
- 2) Using determine resource degradation and perniciousness potentials of lithium battery owed toward the tinny substance-using matter life-cycle effect determination as well as risk determination model in 2020 by Hang yang [23-26].

Re Cellular, the world's biggest cell phone recycling facility, recovered sixteen cell phone batteries. Every year processes nearly 2 million worn phones. The battery recognized 3 types of present battery model that have been determined toward be there the large amount adequate in electronic scrap: Lithium-ion and Li-poly for conventional phone, and dry-cell starting extra experienced smart phones in 2019 by Ogunseitan et al. [27-32]. The California Department of dangerous Substances Control used TCLP, WET, and TLC to classify lithium batteries as solid waste and to identify particular synthetic substances near within quantities to exceeded rigid restrictions in 2018 by Jenny Nelson et al. [33-39]. These procedures were chosen because they were more stringent than other environmental consent procedures in the European Union, Japan, and China in 2019 by Xuwen Luo et al. [16,40,41,55]. If the total concentration of any regulated metal in

the waste removal is equal to or greater than the soluble Threshold Limit Concentration (STLC), use WET to overcome.

## 2. Dangerous waste potential

These results of the metal filtrate test. The results indicate that due to the high level of Co, Cu, as well as, within certain cases, nickel, all dry-cell be supposed to exist confidential as dangerous waste within California law. These findings add to the case for including lithium batteries in California's worldwide Waste rule, which covers seven dangerous waste categories that are frequently produced by households in 2020 by V. Chandra et al. [16,17,22,42-46]. Due to excessive quantities of lead, such Lithium-ion batteries could be there listed as dangerous waste under US regulations, according to the TCLP findings (Pb). Furthermore, some batteries used in the WET procedure exceed the chromium regulatory limits.

According to previous studies on mobile phones without batteries. But, according to the current study, 12.5 percent of cell phone batteries exceed the regulatory threshold in 2009 by Osibanjo O et al. [18].

## 3. Metallic content of Li-on batteries

High levels of aluminum (ranging 51,800 to 3,41,000 mg/Kg), Co (range starting 58,000 toward 2,78,000 mg/Kg), Cu (range starting 54,100 toward 1,52,000 mg/Kg), and lithium (range starting 9,800 toward 37,200 mg/Kg) were found in the TLC results, which were recorded as milligrams of separate metal per kilograms of total battery content (Table 1). The above 4 metals account for 90% percent of the full number of metals on average. Nickel levels, manganese levels, and iron levels be a large amount lesser as well as differed greatly depending on even if these electrode content in 2020 by Yiming Lai [16,21,37,46-51], Table 2.

These were among the other metals found at very low levels. None of the tested lithium batteries contained selenium and so on in 2019 by Chao Peng et al. [17,41,50,52-56]. The combined weight of these metals' accounts for nearly partially about these overall dry-cell mass, through these anode content, electrolytes, diaphragm, and plastic housing accounting for the remainder.

**Table 1**  
Metallic Leachates from Lithium Batteries According to the TTLC Procedure<sup>a,b,c</sup>.

| Element  | TTLC threshold | Li-ion-1 | Li-ion-2 | Li-ion-3 | Li-poly-1 | Li-poly-2 | Li-poly-3 | smart-1 | smart-2 | smart-3 |
|----------|----------------|----------|----------|----------|-----------|-----------|-----------|---------|---------|---------|
| Aluminum | N/A            | 341,000  | 62,300   | 199,000  | 53,870    | 51,800    | 111,000   | 83,700  | 83,210  | 64,300  |
| Cobalt   | 8000           | 140,000  | 58,000   | 667,000  | 220,000   | 203,000   | 164,000   | 241,000 | 278,000 | 165,000 |
| Copper   | 2500           | 95,600   | 58,400   | 95,500   | 152,000   | 123,000   | 76,600    | 96,300  | 54,100  | 75,900  |
| Lithium  | N/A            | 16,900   | 13,900   | 9800     | 29,900    | 31,200    | 19,300    | 34,200  | 37,200  | 26,400  |

<sup>a</sup> Cellular Phone Battery ID (Li-ion = lithium-ion, Li-poly = lithium Polymer, smart = smartphone battery). <sup>b</sup>Inductively coupled plasma mass spectroscopy.

<sup>c</sup> N/A = not applicable.

**Table 2**  
Average Total Potential, within each battery Type, with One Standard Deviation, for All resource Depletion Potential, Human Toxicity Potential, and Exotoxicity Potentials Methods with Life- Cycle Impact assessment (LCIA) – Based Potentials Separated into Impact by Emission (Em.) to Air, Water and soil.

|                                                                     |                                        |                          |                          |                          |                          |                          |
|---------------------------------------------------------------------|----------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Fresh<br>-water<br>Ecotox<br>-icity (kg 1,4-DCB-eq)<br>(LCIA-based) | TRACI Em water<br>(CTUh)               | 3.15 × 10 <sup>-08</sup> | 1.12 × 10 <sup>-08</sup> | 1.86 × 10 <sup>-08</sup> | 7.65 × 10 <sup>-08</sup> | 3.17 × 10 <sup>-08</sup> |
|                                                                     | TRACI Em soil<br>(CTUh)                | 5.00 × 10 <sup>-7</sup>  | 2.78 × 10 <sup>-07</sup> | 1.87 × 10 <sup>-08</sup> | 6.73 × 10 <sup>-08</sup> | 2.72 × 10 <sup>-07</sup> |
|                                                                     | CML Em. Air                            | 2.63 × 10 <sup>01</sup>  | 7.08 × 10 <sup>-01</sup> | 3.15 × 10 <sup>00</sup>  | 9.90 × 10 <sup>-01</sup> | 4.17 × 10 <sup>00</sup>  |
|                                                                     | CML Em. Water<br>(Kg 1,4-DCB-eq)       | 1.40 × 10 <sup>01</sup>  | 3.78 × 10 <sup>00</sup>  | 1.68 × 10 <sup>01</sup>  | 5.30 × 10 <sup>00</sup>  | 2.22 × 10 <sup>01</sup>  |
|                                                                     | CML Em. Soil<br>(Kg 1,4-DCB-eq)        | 8.11 × 10 <sup>01</sup>  | 1.43 × 10 <sup>01</sup>  | 7.78 × 10 <sup>01</sup>  | 3.46 × 10 <sup>01</sup>  | 7.51 × 10 <sup>01</sup>  |
|                                                                     | TRACI Em. Air<br>(CTU <sub>e</sub> )   | 6.42 × 10 <sup>01</sup>  | 1.14 × 10 <sup>01</sup>  | 6.1 × 10 <sup>01</sup>   | 2.74 × 10 <sup>01</sup>  | 5.95 × 10 <sup>01</sup>  |
|                                                                     | TRACI Em. Water<br>(CTU <sub>e</sub> ) | 1.54 × 10 <sup>02</sup>  | 2.71 × 10 <sup>01</sup>  | 1.48 × 10 <sup>02</sup>  | 6.55 × 10 <sup>01</sup>  | 1.42 × 10 <sup>02</sup>  |
|                                                                     | TRACI Em. Soil                         | 8.11 × 10 <sup>01</sup>  | 1.43 × 10 <sup>01</sup>  | 7.78 × 10 <sup>01</sup>  | 3.46 × 10 <sup>01</sup>  | 7.51 × 10 <sup>01</sup>  |
|                                                                     | CML Em.                                | 3.81 × 10 <sup>-01</sup> | 1.11 × 10 <sup>-01</sup> | 4.73 × 10 <sup>-01</sup> | 1.36 × 10 <sup>-01</sup> | 6.55 × 10 <sup>-01</sup> |
|                                                                     | Air (Kg 1,4-DCB-ecotoxicity Eq)        | 8.40 × 10 <sup>-21</sup> | 2.88 × 10 <sup>-21</sup> | 1.11 × 10 <sup>-20</sup> | 3.29 × 10 <sup>-21</sup> | 1.56 × 10 <sup>-20</sup> |
| Terrestrial<br>(LCIA-Based)                                         | CML Em. Water<br>(Kg 1,4-DCB-eq)       | 8.40 × 10 <sup>-21</sup> | 2.88 × 10 <sup>-21</sup> | 1.11 × 10 <sup>-20</sup> | 3.29 × 10 <sup>-21</sup> | 1.56 × 10 <sup>-20</sup> |
|                                                                     | CML Em. Soil<br>(Kg 1,4-DCB-eq)        | 7.63 × 10 <sup>-01</sup> | 2.22 × 10 <sup>-01</sup> | 9.46 × 10 <sup>-06</sup> | 2.72 × 10 <sup>-01</sup> | 2.06 × 10 <sup>-01</sup> |

<sup>a</sup>ELU = Environmental Load Units.

<sup>b</sup>Kg Sb-eq = Kg Antimony equivalent.

<sup>c</sup>TPI = Toxicological Potential Indicator Units.

<sup>d</sup>m3 = inverse of the occupational exposure limit.

<sup>e</sup>CTUh = comparative toxicity units(human toxicity).

<sup>f</sup>Kg 1,4-DCB-eq = Kg 1,4- dichlorobenzene equivalent.

<sup>g</sup>CTUe = comparative toxicity units (ecotoxicity).

#### 4. Supply Depletion, human being harmfulness and toxicological

These CML 2001 and EPS 2000 methods were used to determine biotic resource deficiency potentials, and these methods were used to determine danger human being harmfulness. These CML, as well as TRACI methods, were used to assess the potential for human being harmfulness from effluent to atmosphere, H<sub>2</sub>O, and earth in 2020 by George L. Sun et al. [56–62]. This CML approach was used to determine freshwater harmfulness. For comparison, for all supply depletion perspective, human being harmfulness prospective, and toxicological perspective method are shown in table 2. About average whole prospective, and harmfulness likely used for every metal for all battery forms in 2020 by Young -Tae JO et al. [63–68].

The findings of this study suggest that electrically dry batteries related through manageable electronic devices be the prospective source dangerous metal contaminants in the atmosphere and human health, especially in areas where solid waste collection, processing, and recycling infrastructure is lacking in 2017 by Weifang Gao et al. [62,63,68–74]. Under assumed landfill conditions, metals such as Co, Cu, Ni, and Pb would leach out quantities to would go over legal restrictions, making their relevant dry battery unsafe in central and state law in about the United States in 2018

by Jialing Zhang et al. [75–80]. These consequences necessitate greater policy collaboration to help the recovery of portable electrical batteries.

#### 5. Conclusion

Heavy metals have been showing to be dangerous to equally humans and the environment. These chemicals should be matter to compulsory observation due to their harmfulness and potential for bioaccumulation. For laboratories that analyze a high number of biological or environmental trials daily, numerous operative separation and discovery methods are available. In accumulation, the quick development of molecular biological methods is only if substantial welfares to the investigative sector. Governments should embolden the use of indicators and maintenance unified data collection, study, rule, and conventions. Every one of the two estimation methods designated directly above offers valued evidence that can be used to improve criteria and recommendation beliefs to safeguard human and environmental health from heavy metal pollutants. For the welfare of high-risk populations and subdivisions, revelation data are critical. Additionally, whereas beginning conventional levels or values for chemicals, governments should study the probability for increased acquaintance or exposure of children. Lithium-ion batteries (LIBs) have become one of

the most common energy storage alternatives. The industry has adapted and commercialised many novel materials to meet market demand for these products, including lithium-ion battery materials. Libs' application sectors and market share have exploded in recent years and are continuing to grow. Researchers may be unaware of how LIBs are made and how different processes affect cost, energy consumption, and throughput. We introduce cutting-edge manufacturing technology and examine cost, throughput, and energy consumption based on production processes in a perspective paper. In this perspective research, we look at cutting-edge manufacturing technologies and analyse cost, throughput, and energy usage depending on production methods. We also look at how far the research has progressed, with a focus on the high-cost, energy-intensive, and time-consuming aspects of LIB manufacture.

### CRediT authorship contribution statement

**Sahil Randhawa:** Writing – original draft. **Lalita Chopra:** Writing – review & editing.

### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Acknowledgements

In all humility and gratitude, I am overwhelmed to express my gratitude to all those who have assisted me in putting these concepts, much above the level of simplicity, into something solid. I'd want to express my heartfelt gratitude to my teacher and Chandigarh University for providing me with the excellent opportunity to work on this fantastic research (Pestilential impacts of battery industry discharged metal waste on human health). This also aided me in conducting extensive research and learning about a variety of new topics. I owe them a debt of gratitude. Without the support and guidance of teachers, friends, no endeavor at any level can be done satisfactorily.

### References

- [1] D.H.P. Kang, M. Chen, O.A. Ogunseitan, Potential environmental and human health impacts of rechargeable lithium batteries in electronic waste, *Environmental science & technology* 47 (10) (2013) 5495–5503.
- [2] J. Jegan Roy, M. Srinivasan, B. Cao, Bioleaching as an Eco-Friendly Approach for Metal Recovery from Spent NMC-Based Lithium-Ion Batteries at a High Pulp Density, *ACS Sustainable Chemistry & Engineering* 9 (8) (2021) 3060–3069.
- [3] O.A. Ogunseitan, J.M. Schoenung, J.-D.- M. Saphores, A.A. Shapiro, The electronics revolution: from e-wonderland to e-wasteland, *Science* 326 (5953) (2009) 670–671.
- [4] J. Xu, H.R. Thomas, R.W. Francis, K.R. Lum, J. Wang, B.o. Liang, A review of processes and technologies for the recycling of lithium-ion secondary batteries, *Journal of Power Sources* 177 (2) (2008) 512–527.
- [5] R.J. Hamers, Energy Storage Materials as Emerging Nano-contaminants, *Chemical research in toxicology* 33 (5) (2020) 1074–1081.
- [6] J.M. Schoenung, O.A. Ogunseitan, J.D.M. Saphores, A.A. Shapiro, Adopting lead-free electronics: policy differences and knowledge gaps, *Journal of Industrial Ecology* 8 (4) (2004) 59–85.
- [7] J.D. Lincoln, O.A. Ogunseitan, A.A. Shapiro, J.-D. Saphores, Leaching assessments of hazardous materials in cellular telephones, *Environmental Science & Technology* 41 (7) (2007) 2572–2578.
- [8] S.E. Musson, K.N. Vann, Y.-C. Jang, S. Mutha, A. Jordan, B. Pearson, T.G. Townsend, RCRA toxicity characterization of discarded electronic devices, *Environmental science & technology* 40 (8) (2006) 2721–2726.
- [9] I.C. Nnorom, O. Osibanjo, Heavy metal characterization of waste portable rechargeable batteries used in mobile phones, *International Journal of Environmental Science & Technology* 6 (4) (2009) 641–650.
- [10] L. Gaines, J. Sullivan, A. Burnham, I. Belharouak, Life-cycle analysis of production and recycling of lithium ion batteries, *Transportation research record* 2252 (1) (2011) 57–65.
- [11] G. Majeau-Bettez, T.R. Hawkins, A.H. Strømman, Life cycle environmental assessment of lithiumion and nickel metal hydride batteries for plug-in hybrid and battery electric vehicles, *Environmental science & technology* 45 (10) (2011) 4548–4554.
- [12] D.A. Notter, M. Gauch, R. Widmer, P. Wager, A. Stamp, R. Zah, H.J. Althaus, Contribution of Li-ion batteries to the environmental impact of electric vehicles, 2010, pp.68–71.
- [13] P. Van den Bossche, F. Vergels, J. Van Mierlo, J. Matheys, W. Van Autenboer, SUBAT: An assessment of sustainable battery technology, *Journal of power sources* 162 (2) (2006) 913–919.
- [14] A.L.S. Global, USEPA Method 1311–Toxicity Characteristic Leaching Procedure (TCLP), 2011, pp.78–85.
- [15] W. Gao, X. Zhang, X. Zheng, X. Lin, H. Cao, Y.i. Zhang, Z. Sun, Lithium carbonate recovery from cathode scrap of spent lithium-ion battery: a closed-loop process, *Environmental science & technology* 51 (3) (2017) 1662–1669.
- [16] P. Yadav, C.J. Jie, S. Tan, M. Srinivasan, Recycling of cathode from spent lithium iron phosphate batteries, *Journal of Hazardous Materials* 399 (2020) 123068, <https://doi.org/10.1016/j.jhazmat.2020.123068>.
- [17] H. Pinegar, Y.R. Smith, M. Beneficiation, of, End-of-Life, Lithium-Ion, Battery, Components., In, Energy, Technology., Recycling, Carbon Dioxide Management, and Other Technologies 2020 (2020) 259–267.
- [18] Kevin N. Vann, Stephen E. Musson, Timothy G. Townsend, Factors affecting TCLP lead leach ability from computer CPUs, *Waste Management* 26 (3) (2006) 293–298.
- [19] J.E. Hermansen, T.L.T. Nguyen, Life cycle assessment and the agri-food chain, In *Green Technologies in Food Production and Processing* (2012) 43–60.
- [20] Seong-Rin Lim, Daniel Kang, Oladele A. Ogunseitan, Julie M. Schoenung, Potential environmental impacts from the metals in incandescent, compact fluorescent lamp (CFL), and light-emitting diode (LED) bulbs, *Environmental science & technology* 47 (2) (2013) 1040–1047.
- [21] Weronika Urbańska, Magdalena Osial, Investigation of the Physico-Chemical Properties of the Products Obtained after Mixed Organic-Inorganic Leaching of Spent Li-Ion Batteries, *Energies* 13 (24) (2020) 6732, <https://doi.org/10.3390/en13246732>.
- [22] Theoneste Nshizirungu, Ashutosh Agarwal, Young Tae Jo, Masud Rana, Dong Shin, Jeong-Hun Park, Chlorinated polyvinyl chloride (CPVC) assisted leaching of lithium and cobalt from spent lithium-ion battery in subcritical water, *Journal of hazardous materials* 393 (2020) 122367, <https://doi.org/10.1016/j.jhazmat.2020.122367>.
- [23] Tairan Yang, Yingqi Lu, Liurui Li, Dayang Ge, Heng Yang, Weinan Leng, Hui Zhou, Xu Han, Nolan Schmidt, Michael Ellis, Zheng Li, An Effective Relithiation Process for Recycling Lithium - Ion Battery Cathode Materials, *Advanced Sustainable Systems* 4 (1) (2020) 1900088, <https://doi.org/10.1002/adsu.v4.1.10.1002/adsu.201900088>.
- [24] Z. Loukzadeh, S.A. Sharifian, O. Aminian, A. Shojaaoddiny-Ardekani, Pulmonary effects of spot welding in automobile assembly, *Occupational Medicine* 59 (4) (2009) 267–269.
- [25] M.M. Wang, C.C. Zhang, F.S. Zhang, F.S., An environmental benign process for cobalt and lithium recovery from spent lithium-ion batteries by mechanochemical approach, *Waste management* (2016) 239–244.
- [26] Xianlai Zeng, Jinhui Li, Innovative application of ionic liquid to separate Al and cathode materials from spent high-power lithium-ion batteries, *Journal of hazardous materials* 271 (2014) 50–56.
- [27] Narendra Singh, Huabo Duan, Oladele A. Ogunseitan, Jinhui Li, Yuanyuan Tang, Toxicity trends in E-Waste: A comparative analysis of metals in discarded mobile phones, *Journal of hazardous materials* 380 (2019) 120898, <https://doi.org/10.1016/j.jhazmat.2019.120898>.
- [28] Vicharana Intrakamhaeng, Kyle A. Clavier, Timothy G. Townsend, Initiatives to reduce lead from electronic devices: evidence of success from the toxicity characteristic leaching procedure, *Journal of the Air & Waste Management Association* 69 (9) (2019) 1116–1121.
- [29] Haruka Pinegar, York R. Smith, Recycling of end-of-life lithium ion batteries, part I: commercial processes, *Journal of Sustainable Metallurgy* 5 (3) (2019) 402–416.
- [30] Haikal Ismail, Marlia M. Hanafiah, An overview of LCA application in WEEE management: Current practices, progress and challenges, *Journal of Cleaner Production* 232 (2019) 79–93.
- [31] Bhumika R. Khatri, Devayani R. Tipre, Shailesh R. Dave, Comparison of Hydro-and Bio hydrometallurgical extraction of metals from waste li-ion batteries of cell phone, *Journal of Sustainable Metallurgy* 5 (2) (2019) 250–261.
- [32] Chang-Heum Jo, Seung-Taek Myung, Efficient recycling of valuable resources from discarded lithium-ion batteries, *Journal of Power Sources* 426 (2019) 259–265.
- [33] Violaine Sironval, Laurence Reylandt, Perrine Chaurand, Saloua Ibouaadataen, Mihaly Palmi-Pallag, Yousof Yakoub, Bernard Ucakar, Jérôme Rose, Claude Poleunis, Rita Vanbever, Etienne Marbaix, Dominique Lison, Sybille van den Brule, Respiratory hazard of Li-ion battery components: elective toxicity of lithium cobalt oxide (LiCoO<sub>2</sub>) particles in a mouse bioassay, *Archives of toxicology* 92 (5) (2018) 1673–1684.
- [34] Xiangping Chen, Chunxiu Guo, Hongrui Ma, Jiazhu Li, Tao Zhou, Ling Cao, Duozihi Kang, Organic reductants based leaching: A sustainable process for the recovery of valuable metals from spent lithium ion batteries, *Waste Management* 75 (2018) 459–468.
- [35] Sheridan Few, Oliver Schmidt, Gregory J. Offer, Nigel Brandon, Jenny Nelson, Ajay Gambhir, Prospective improvements in cost and cycle life of off-grid



- lithium-ion battery packs: An analysis informed by expert elicitations, *Energy Policy* 114 (2018) 578–590.
- [36] Kevin M. Winslow, Steven J. Laux, Timothy G. Townsend, A review on the growing concern and potential management strategies of waste lithium-ion batteries, *Resources, Conservation and Recycling* 129 (2018) 263–277.
- [37] Haruka Pinegar, York R. Smith, End-of-life lithium-ion battery component mechanical liberation and separation, *JOM* 71 (12) (2019) 4447–4456.
- [38] Meng-Meng Wang, Cong-Cong Zhang, Fu-Shen Zhang, An environmental benign process for cobalt and lithium recovery from spent lithium-ion batteries by mechanochemical approach, *Waste management* 51 (2016) 239–244.
- [39] M. Xue, Z. Xu, Application of life cycle assessment on electronic waste management: a review, *Environmental management*, 2009, pp.693–707.
- [40] Juan Liu, Xuwen Luo, Yuqing Sun, Daniel C.W. Tsang, Jianying Qi, Weilong Zhang, Nuo Li, Meiling Yin, Jin Wang, Holger Lippold, Yongheng Chen, Guodong Sheng, Thallium pollution in China and removal technologies for waters: a review, *Environment international* 126 (2019) 771–790.
- [41] F. Gu, J. Guo, X. Yao, P.A. Summers, S.D. Widiyatmoko, P. Hall, P., An investigation of the current status of recycling spent lithium-ion batteries from consumer electronics in China, *Journal of cleaner production* (2017) 765–780.
- [42] Roberto Baca, Reliable energy strategy based on incandescent lighting, *Energy and Power, Engineering* 07 (05) (2015) 174–180.
- [43] V. Chandran, A. Ghosh, C.K. Patil, V. Mohanavel, A.K. Priya, R. Rahim, R. Madavan, U. Muthuraman, A. Karthick, Comprehensive review on recycling of spent lithium-ion batteries, *Materials Today: Proceedings* (2021) 256–265.
- [44] Fupeng Liu, Chao Peng, Quanxin Ma, Jinliang Wang, Songlin Zhou, Zaoming Chen, Benjamin P. Wilson, Mari Lundström, Selective lithium recovery and integrated preparation of high-purity lithium hydroxide products from spent lithium-ion batteries, *Separation and Purification Technology* 259 (2021) 118181, <https://doi.org/10.1016/j.seppur.2020.118181>.
- [45] Saptak Rarotra, Satyabrata Sahu, Pawan Kumar, Ki-Hyun Kim, Yiu Fai Tsang, Vikas Kumar, Parveen Kumar, Madhavi Srinivasan, Andrei Veksha, Grzegorz Lisak, Progress and challenges on battery waste management: A critical review, *Chemistry Select* 5 (20) (2020) 6182–6193.
- [46] Oleksandr Dolotko, Ihor Z. Hlova, Yaroslav Mudryk, Shalabh Gupta, Viktor P. Balema, Mechanochemical recovery of Co and Li from LCO cathode of lithium-ion battery, *Journal of Alloys and Compounds* 824 (2020) 153876, <https://doi.org/10.1016/j.jallcom.2020.153876>.
- [47] Jun Li, Yiming Lai, Xianqing Zhu, Qiang Liao, Ao Xia, Yun Huang, Xun Zhu, Pyrolysis kinetics and reaction mechanism of the electrode materials during the spent LiCoO<sub>2</sub> batteries recovery process, *Journal of Hazardous Materials* 398 (2020) 122955, <https://doi.org/10.1016/j.jhazmat.2020.122955>.
- [48] Siqi Zhao, Wenxuan Zhang, Guangming Li, Haochen Zhu, Juwen Huang, Wenzhi He, Ultrasonic renovation mechanism of spent LCO batteries: A mild condition for cathode materials recycling, *Resources, Conservation and Recycling* 162 (2020) 105019, <https://doi.org/10.1016/j.resconrec.2020.105019>.
- [49] George L. Sun, Erin E. Reynolds, Angela M. Belcher, Using yeast to sustainably remediate and extract heavy metals from waste waters, *Nature Sustainability* 3 (4) (2020) 303–311.
- [50] Junjie Shi, Chao Peng, Min Chen, Yun Li, Hurman Eric, Lassi Klemettinen, Mari Lundström, Pekka Taskinen, Ari Jokilaakso, Sulfation Roasting Mechanism for Spent Lithium-Ion Battery Metal Oxides Under SO<sub>2</sub>-O<sub>2</sub>-Ar Atmosphere, *JOM* 71 (12) (2019) 4473–4482.
- [51] Pratima Meshram, Abhilash Mishra, Abhilash, Rina Sahu, Environmental impact of spent lithium ion batteries and green recycling perspectives by organic acids—A review, *Chemosphere* 242 (2020) 125291, <https://doi.org/10.1016/j.chemosphere.2019.125291>.
- [52] Clifton Westly Evers, Polluted leisure, *Leisure Sciences* 41 (5) (2019) 423–440.
- [53] M.N. Haji, A.H. Slocum, An offshore solution to cobalt shortages via adsorption-based harvesting from seawater, *Renewable and Sustainable Energy Reviews* (2019) 301–309.
- [54] S.R. Lim, J.M. Schoenung, Human health and ecological toxicity potentials due to heavy metal content in waste electronic devices with flat panel displays, *Journal of Hazardous Materials* (2010) 251–259.
- [55] L.V. Juan, Mingxing Zhang, Miqi Tang, Qi Lu, Yufei Qin, Yan Lu, Bo Yu, Recycling lithium cobalt oxide from its spent batteries: An electrochemical approach combining extraction and synthesis, *Journal of Hazardous Materials* (2021) 251–259.
- [56] F. Liu, C. Peng, Q. Ma, J. Wang, S. Zhou, Z. Chen, B.P. Wilson, M. Lundström, Selective lithium recovery and integrated preparation of high-purity lithium hydroxide products from spent lithium-ion batteries, *Separation and Purification Technology* (2020) 118181.
- [57] J.S. Ribeiro, M.B. Freitas, J.C. Freitas, Recycling of graphite and metals from spent Li-ion batteries aiming the production of graphene/CoO-based electrochemical sensors, *Journal of Environmental, Chemical Engineering* (2021) 104689.
- [58] X. Yuan, X. Li, Vanadium Hexacyanoferrate Derived V-Fe-K Mixed Oxides as Anode Materials for Lithium-Ion Batteries, *Chemistry Select* (2020) 13748–13753.
- [59] H. Rallo, L.C. Casals, D. De La Torre, R. Reinhardt, C. Marchante, B. Amante, Lithium-ion battery 2nd life used as a stationary energy storage system: Ageing and economic analysis in two real cases, *Journal of Cleaner Production* (2020) 122584.
- [60] X. Zeng, J. Li, Innovative application of ionic liquid to separate Al and cathode materials from spent high-power lithium-ion batteries, *Journal of hazardous materials* (2020) 50–56.
- [61] T. Nshizirungu, M. Rana, Y.T. Jo, J.H. Park, Rapid leaching and recovery of valuable metals from spent lithium ion batteries (LIBs) via environmentally benign subcritical nickel-containing water over chlorinated polyvinyl chloride, *Journal of hazardous materials* (2020) 122667.
- [62] W. Gao, X. Zhang, X. Zheng, X. Lin, H. Cao, Y. Zhang, Z.H.I. Sun, Lithium carbonate recovery from cathode scrap of spent lithium-ion battery: a closed-loop process, *Environmental science & technology* (2017) 1662–1669.
- [63] S. Sun, C. Jin, W. He, G. Li, H. Zhu, J. Huang, Management status of waste lithium-ion batteries in China and a complete closed-circuit recycling process, *Science of The Total Environment* (2021) 145913.
- [64] K. He, Z.Y. Zhang, F.S. Zhang, A green process for phosphorus recovery from spent LiFePO<sub>4</sub> batteries by transformation of delithiated LiFePO<sub>4</sub> crystal into NaFeS<sub>2</sub>, *Journal of hazardous materials* (2020) 122614.
- [65] M. Chen, Z. Zheng, Q. Wang, Y. Zhang, X. Ma, C. Shen, D. Xu, J. Liu, P. Liu, I. O'Connor Gionet, Closed loop recycling of electric vehicle batteries to enable ultra-high quality cathode powder, *Scientific reports* (2019) 1–9.
- [66] G. Li, L. An, Impacts of Recycling of Spent Lithium-Ion Batteries on Environmental Burdens, In *Recycling of Spent Lithium-Ion Batteries* (2019) 199–217.
- [67] S. Bashir, N. King Sanders, J.L. Liu, Technology Policy and Road Map of Battery, In *Nanostructured Materials for Next-Generation Energy Storage and Conversion* (2019) 1–59.
- [68] J. Zhang, J. Hu, W. Zhang, Y. Chen, C. Wang, Efficient and economical recovery of lithium, cobalt, nickel, manganese from cathode scrap of spent lithium-ion batteries, *Journal of cleaner production* (2018) 437–446.
- [69] Y. Chen, M. Chen, Y. Li, S. Chen, Z. Xu, Impact of technological innovation and regulation development on e-waste toxicity: a case study of waste mobile phones, *Scientific reports* (2018) 1–9.
- [70] P. Pathak, D.N. Singh, G.G. Pandit, P.R. Apte, Establishing sensitivity of distribution coefficient on various attributes of a soil-contaminant system, *Journal of Hazardous, Toxic, and Radioactive Waste* (2014) 64–75.
- [71] G. Zubi, R. Dufo-Lopez, M. Carvalho, G. Pasaoglu, The lithium-ion battery: State of the art and future perspectives, *Renewable and Sustainable Energy Reviews* (2018) 292–308.
- [72] L.P. He, S.Y. Sun, J.G. Yu, Performance of LiNi<sub>1/3</sub>Co<sub>1/3</sub>Mn<sub>1/3</sub>O<sub>2</sub> prepared from spent lithiumion batteries by a carbonate co-precipitation method, *Ceramics International* (2018) 351–357.
- [73] M. Zackrisson, K. Fransson, J. Hildenbrand, G. Lampic, C. O'Dwyer, Life cycle assessment of lithium-air battery cells, *Journal of Cleaner Production* (2016) 299–311.
- [74] S. Podder, M.Z.R. Khan, Comparison of lead acid and Li-ion battery in solar home system of Bangladesh, in: In 2016 5th International Conference on Informatics, Electronics and Vision (ICIEV), 2016, pp. 434–438.
- [75] D. Larcher, J.M. Tarascon, Towards greener and more sustainable batteries for electrical energy storage, *Nature chemistry* (2015) 19–29.
- [76] E.N. Keyzer, J. Lee, Z. Liu, A.D. Bond, D.S. Wright, C.P. Grey, A general synthetic methodology to access magnesium aluminates electrolyte systems for Mg batteries, *Journal of Materials Chemistry A* (2019) 2677–2685.
- [77] A. Reizabal, R. Gonçalves, A. Fidalgo-Marijuan, C.M. Costa, L. Perez, J.L. Vilas, S. Lancers-Mendez, Tailoring silk fibroin separator membranes pore size for improving performance of lithium ion batteries, *Journal of Membrane Science* (2020) 117678.
- [78] M. Sethurajan, S. Gaydardzhiev, Bio processing of spent lithium ion batteries for critical metals recovery—A review, *Resources, Conservation and Recycling* (2021) 105225.
- [79] D. Purchase, G. Abbasi, L. Bisschop, D. Chatterjee, C. Ekberg, M. Ermolin, P. Fedotov, N.G. Kandile, M. Lundstrom, The waste stream of obsolete electronic equipment, *Pure and Applied Chemistry* (2020) 256–266.
- [80] A. Bilich, K. Langham, R. Geyer, L. Goyal, J. Hansen, A. Krishnan, J. Bergesen, P. Sinha, Life cycle assessment of solar photovoltaic micro grid systems in off-grid communities, *Environmental science & technology* (2017) 1043–1052.